HOBIT: Hardness Optimized Batch Sampling for InfoNCE Training

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Abstract

Contrastive training with InfoNCE loss and in-batch negatives is the standard approach for learning dual-encoder models. Its effectiveness, however, critically depends on the availability of hard negatives; in their absence, learning quickly saturates. Existing methods address this via explicit hard-negative mining, which is often costly or heuristic-driven. We introduce **HOBIT**, a principled mini-batch construction method that improves in-batch negative quality by reordering training examples at every epoch. HOBIT solves an optimization problem motivated by the InfoNCE objective to yield mini-batches such that each query in the batch is exposed to hard yet non-contradictory, informative negative examples. We show that the optimization objective is monotone and submodular which in turn leads us to a greedy algorithm that admits the standard $\mathcal{O}(1-1/e)$ approximation guarantee. Empirically, we show that HOBIT incurs negligible computational overhead while significantly outperforming state-of-the-art batching methods, and remains complementary to existing hard negative mining techniques.[1]

1. Introduction

Dual encoders are a prevalent architecture for building large-scale information retrieval systems. They are embedding models that encode queries and documents independently, and are trained using contrastive learning to bring representations of relevant query–document pairs closer than irrelevant ones. The contrastive loss used for training depends on the supervision available in the data. When explicit labels for both positive and negative documents are provided for each query, one can directly optimize losses (Gupta et al., 2024b; Robinson et al., 2021) that exploit this richer supervision.

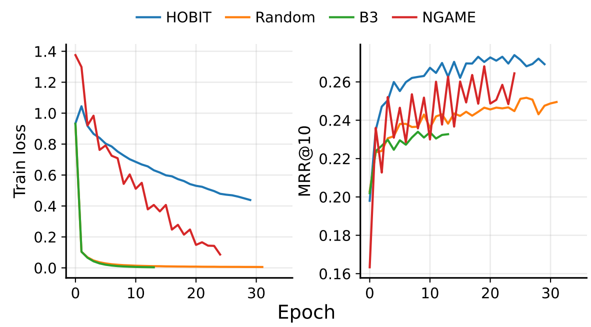

Figure 1. Training dynamics on MSMARCO illustrating the importance of hard negatives in InfoNCE training. Left: epoch-wise training loss. Right: validation MRR@10. Methods that rely on static or weakly adaptive batch construction (Random, B3) saturate early, while adaptive hard-negative batching methods (NGAME and HOBIT) sustain learning and achieve better generalization.

However, real-world retrieval datasets provide supervision only for positive query–document pairs, offering little to no guidance on what constitutes a meaningful negative.

In this weakly supervised regime, training typically relies on InfoNCE-style losses with in-batch negatives (Karpukhin et al., 2020; Qu et al., 2021; Gao & Callan, 2021; 2022), where positives of other queries within the same mini-batch are treated as negatives. While effective, this paradigm succeeds only when the in-batch negatives are sufficiently hard (Lin et al., 2021a; Zhan et al., 2021; Robinson et al., 2021); when batches are dominated by easy or redundant negatives, learning quickly saturates (see Fig. 1).

A large body of prior work has therefore focused on hard-negative mining. Most existing approaches either rely on expensive multi-stage training pipelines (Xiong et al., 2021; Sun et al., 2022) or select negatives using heuristic criteria (Dahiya et al., 2023; Yang et al., 2023). A common design pattern is to maintain an approximate nearest-neighbor index over document embeddings and query it during training to retrieve hard negatives. However, because document representations evolve continuously as the model trains, such indices quickly become stale and require frequent rebuilding to remain aligned with the current model state.

At scale, this indexing cost becomes prohibitive. For example, MSMARCO (Nguyen et al., 2016) roughly contains only 500k query–positive pairs compared to 8 million corpus documents, making per-epoch document index recon-

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[1]Anonymized code provided as supplementary material.

struction impractical. As a result, most methods rely on static or stale indices. Approaches such as ANCE (Xiong et al., 2021) partially mitigate this by refreshing the index every few epochs. However, recent study (Monath et al., 2023) shows that stale negative mining biases gradient updates, effectively optimizing a surrogate objective that can significantly diverge from the true training objective. Consequently, balancing computational efficiency with training correctness remains a key challenge in hard-negative mining.

In this paper, we step back and ask a simpler question: *Can the batch itself be made informative?* We introduce **HOBIT**, a lightweight and principled batching strategy that deliberately constructs mini-batches so that examples serve as hard and informative negatives for one another. For MSMARCO, HOBIT optimizes over the 500k (q, d) training pairs, making it orders of magnitude cheaper than approaches like ANCE, which must search over the entire 8 million–document corpus for each query in the batch.

By analyzing the gradient of the InfoNCE loss with respect to a query embedding $\vec{q}_i$, we show that an informative mini-batch should include negative documents $\vec{d}_j$ (paired with other queries in the batch) that satisfy two complementary properties relative to the positive document $\vec{d}_i$ of query q_i. Specifically, such negatives should be (i) *hard*, exhibiting high query–document similarity ($\vec{q}_i^\top \vec{d}_j \uparrow$), while remaining (ii) *non-contradictory*, i.e., well separated from the positive document ($\vec{d}_i^\top \vec{d}_j \downarrow$). Selecting examples that jointly satisfy these criteria across a mini-batch leads to a combinatorial optimization problem over a large discrete space, which is NP-hard. We show, however, that HOBIT induces a monotone submodular objective, enabling an efficient greedy optimization algorithm with a $(1 - 1/e)$ approximation guarantee. Our theoretical analysis further shows that this objective faithfully reflects the training dynamics of InfoNCE, providing a principled link between batch construction and contrastive learning.

Through extensive experiments, we show that HOBIT consistently outperforms state-of-the-art batching baselines across three datasets and two encoder architectures. On MSMARCO Passage (Nguyen et al., 2016), it improves MRR@10 by $+0.6$ points with `roberta-base` (Liu et al., 2019) and by $+1.5$ points with the stronger `e5-large-unsupervised` encoder (Wang et al., 2022). Gains are larger on Natural Questions (Kwiatkowski et al., 2019), where Recall@10 increases by $+3.3$ and $+1.6$ for the two encoders, respectively. Similar trends hold on TriviaQA (Joshi et al., 2017), with Recall@10 improvements of $+1.0$ and $+1.7$. Beyond outperforming standalone batching strategies, HOBIT complements explicit hard-negative mining, remains effective under sparse relevance supervision, and admits a cached variant, HOBIT-C, that achieves up to 29% per-epoch training speedups with negligible impact on retrieval quality, making principled batch construction practical at scale.

Contributions. Our main contributions are:

1. We propose HOBIT, a principled batching method that yields informative in-batch negatives tailored to the InfoNCE objective.
2. We formalize two complementary criteria, hardness and non-contradiction, that are essential for an effective mini-batch.
3. We further propose HOBIT-C as an efficient approximation to HOBIT and provide formal bounds on its approximation error.
4. We empirically show that HOBIT consistently outperforms state-of-the-art batching methods across datasets and encoders, remains complementary to hard-negative mining, and is robust to sparse relevance supervision while enabling substantial training speedups via caching.

2. Motivating the Need for Hard Negatives

We motivate the need for having in-batch hard negatives using a simple experiment on MSMARCO. Figure 1 plots the training loss and validation performance (MRR@10) across epochs for four batch sampling methods: **Random**, which samples mini-batches uniformly at random; **B3**, which determines a fixed batch order once before training using a teacher model; **NGAME**, which adaptively reorders batches at each epoch based on clustering query embeddings; and **HOBIT**, our method. As shown in the left panel, training with Random and B3 quickly reaches a plateau. In contrast, NGAME and HOBIT, both of which adapt batch construction to the evolving encoder representations, maintain a consistently higher training loss over many epochs. This loss trend reflects the sustained presence of challenging in-batch negatives that continue to provide strong gradient signals to the model. This difference in training dynamics directly translates to the validation performance shown in the right panel. While Random and B3 yield poor validation MRR, adaptive methods, by contrast, achieve substantially better retrieval performance, with HOBIT consistently outperforming NGAME. These observations strongly motivate the need for a principled, training loss-aware mining of negative examples, which we develop in the following sections. We discuss a qualitative example in details in Appendix A.1.

3. Related Work

We review prior work on mini-batching in this section, and defer a detailed discussion on dual-encoder training and hard-negative mining to Appendix A.2.

Batching methods. Batch construction has received far less

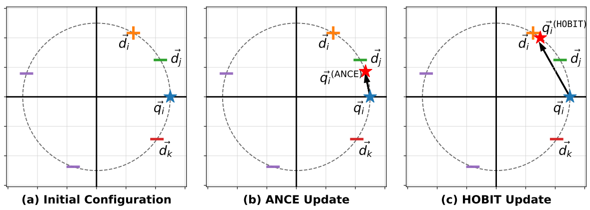

Figure 2. **Effect of negative selection on the query embedding.** All vectors are unit-normalized and lie on the unit circle. **(a)** Initial configuration showing a query $\vec{q}_i$, its positive document $\vec{d}_i$, a near-positive hard negative $\vec{d}_j$, and a distant hard negative $\vec{d}_k$. **(b)** Update induced by selecting the near-positive negative $\vec{d}_j$ using query–document similarity-based hard negative mining, as in ANCE (Xiong et al., 2021). $\vec{q}_i^{\,(\mathrm{ANCE})}$ is the updated query. **(c)** Update induced by selecting the distant negative $\vec{d}_k$ as favored by HOBIT. Selecting negative based on Eq. 6 favours $\vec{d}_k$ over $\vec{d}_j$ for $\alpha \geq 0.5$. $\vec{q}_i^{\,(\mathtt{HOBIT})}$ is the updated query. Updates are performed with the learning rate of 1.0.

attention in the literature compared to hard-negative mining, despite its ability to improve retrieval performance without full-corpus indexing. Most existing methods form batches by clustering queries or query–document interaction embeddings, thereby aiming to increase the hardness of in-batch negatives (Dahiya et al., 2023; Yang et al., 2023; Thirukovalluru et al., 2025). These studies demonstrate that batch composition alone can significantly enhance retrieval quality; however, the resulting batching objectives are largely heuristic-driven and not explicitly tied to the training objective. Complementary work exploits batch structure and interaction graphs to increase the diversity and semantic coherence of in-batch negatives (Lin et al., 2021b; Moiseev et al., 2023; Morris & Rush, 2025; Sachidananda et al., 2023; Thirukovalluru et al., 2025). Collectively, these works position principled batch construction as a lightweight and practical alternative to hard-negative mining.

4. Problem Formulation

Let $\mathcal{Q}$ denote the set of queries and $\mathcal{D}$ the document corpus from which relevant results are retrieved. Our training dataset $D_{\text{trn}} = \{(q_i, d_i)\}_{i=1}^{N}$ comprises N query–positive pairs, where each training instance consists of a query $q_i \in \mathcal{Q}$ and an associated relevant document $d_i \in \mathcal{D}$. When a query has multiple relevant documents, it appears multiple times in D_{trn}, once for each positive.

Our goal is to learn a dual encoder model $f_\theta : \mathcal{Q} \cup \mathcal{D} \to \mathbb{R}^d$ that embeds queries and documents into a shared d-dimensional vector space. The training objective is to ensure that each query is more similar to its relevant documents than to any irrelevant document in the corpus. We assume that f_θ produces unit-normalized embeddings and assess relevance for a q-d pair using cosine similarity. Because $\mathcal{D}$ is large in realistic retrieval datasets, dual encoders are typically trained using the InfoNCE loss with in-batch negative sampling, which we describe next.

InfoNCE Objective. Given a mini-batch $B = \{(q_i, d_i)\}_{i=1}^{b} \subset D_{\text{trn}}$, in-batch contrastive training treats the positive documents associated with other queries in the same batch as negatives. Specifically, for each query q_i, its paired document d_i is the positive example, while the documents $\{d_j\}_{j \neq i}$ in B act as negatives. Let $\vec{q} = f_\theta(q)$ and $\vec{d} = f_\theta(d)$ denote the query and document embeddings, respectively. The similarity for a pair (q_i, d_j) is defined as:

$$s_{ij} = f_\theta(q_i)^\top f_\theta(d_j) = \vec{q}_i^{\,\top} \vec{d}_j. \tag{1}$$

For a temperature parameter $\tau > 0$, per-query InfoNCE loss is

$$\mathcal{L}_i = -\log \frac{\exp(s_{ii}/\tau)}{\sum_{j=1}^{b} \exp(s_{ij}/\tau)} \tag{2}$$

Defining the normalized similarity scores as

$$P_{ij} = \frac{\exp(s_{ij}/\tau)}{\sum_{k=1}^{b} \exp(s_{ik}/\tau)}, \tag{3}$$

the loss in Eq. 2 simplifies to $\mathcal{L}_i = -\log P_{ii}$.

To assess how batch composition influences learning, we begin our analysis by examining the gradient of the InfoNCE loss with respect to the query embedding.

Lemma 4.1 (Gradient w.r.t. query embedding)**.** *For the InfoNCE loss $\mathcal{L}_i$, the gradient with respect to the query embedding $\vec{q}_i$ is given by* $\nabla_{\vec{q}_i} \mathcal{L}_i = \frac{1}{\tau} \left(\sum_{j=1}^{b} P_{ij}\, \vec{d}_j - \vec{d}_i \right)$,

Proof deferred to Appendix A.3.1. □

Remark 4.2. Lemma 4.1 shows that learning is governed by competition between the positive document embedding $\vec{d}_i$ and a softmax-weighted average of batch documents. When a mini-batch contains only easy negatives, i.e., $s_{ii} \gg s_{ij}$ for all $j \neq i$, the softmax collapses to $P_{ii} \approx 1$ and $P_{ij} \approx 0$. As a result, the two terms in the gradient expression nearly cancel, yielding $\nabla_{\vec{q}_i} \mathcal{L}_i \approx 0$ and learning stalls.

Theorem 4.3 (Dominant negative)**.** *Fix a query q_i and suppose the in-batch softmax distribution satisfies*

$$P_{ii} = 1 - p - \epsilon, \qquad P_{ij} = p, \qquad \sum_{k \notin \{i,j\}} P_{ik} = \epsilon, \tag{4}$$

for some $j \neq i$, with $p \in (0,1)$ and $\epsilon \in (0, 1-p)$. Then the gradient of the InfoNCE loss satisfies

$$\left\| \nabla_{\vec{q}_i} \mathcal{L}_i \right\|_2 \;\geq\; \frac{p}{\tau} \sqrt{2(1 - t_{ij})} \;-\; \frac{2\epsilon}{\tau}. \tag{5}$$

where, $t_{ij} = \vec{d}_i^{\,\top} \vec{d}_j$.

Proof deferred to Appendix A.3.2. □

Remark 4.4 (Desiderata for an effective mini-batch). For a training pair (q_j, d_j) in a mini-batch to provide a non-trivial learning signal for the loss incurred by (q_i, d_i) under the InfoNCE objective, the corresponding in-batch negative d_j must satisfy two complementary conditions:

1. *Hard:* it must attract a non-negligible softmax probability mass, i.e., the weight p should be large. This requires the query-negative similarity $s_{ij} = \vec{q_i}^\top \vec{d_j}$ to be high.
2. *Non-contradictory:* it must be well separated from the positive document, meaning the similarity $t_{ij} = \vec{d_i}^\top \vec{d_j}$ should be small. Negatives that are near-duplicates of the positive are ambiguous and produce weak directional gradients, even when they are hard.

Definition 4.5 (Hardness Score)**.** We quantify the usefulness of an in-batch item (q_j, d_j) for another item (q_i, d_i) using the following measure:

$$w_{ij} \;=\; \vec{q_i}^\top \vec{d_j} - \alpha \vec{d_i}^\top \vec{d_j}, \tag{6}$$

where $\alpha > 0$ controls the trade-off between hardness and non-contradiction.

Geometric Motivation for w_{ij}. Figure 2 provides a geometric intuition for the pairwise hardness score w_{ij}. Panel (a) shows a query $\vec{q_i}$, its positive document $\vec{d_i}$, along with two hard negatives $\vec{d_j}$ and $\vec{d_k}$, where $\vec{d_j}$ is closer to the positive document $\vec{d_i}$, while $\vec{d_k}$ is far apart. Assuming a batch size of two, we illustrate how InfoNCE loss updates $\vec{q_i}$ when applied with the documents $\{\vec{d_i}, \vec{d_j}\}$ in panel (b) as opposed to $\{\vec{d_i}, \vec{d_k}\}$ in panel (c). Methods such as ANCE (Xiong et al., 2021), which select negatives based solely on the query–document similarity, tend to prefer negatives like $\vec{d_j}$. As shown in panel (b), such near-positive negatives induce only a weak improvement in the query's projection onto the positive direction $\vec{d_i}$. In contrast, panel (c) shows that the negatives preferred by `HOBIT` induce a substantially larger increase in projection toward $\vec{d_i}$. These findings underscore that informative negatives are not merely those that score highly with respect to the query, but those that also remain well separated from the positive.

5. `HOBIT` Solution Approach

Given a seed set $\mathcal{S} \subset D_{\text{trn}}$ with $|\mathcal{S}| = s < b$, the batch construction problem aims to select $b - s$ additional examples from the candidate pool $\mathcal{U} = D_{\text{trn}} \setminus \mathcal{S}$ to form a mini-batch $B = \mathcal{S} \cup X$. The objective is to choose X so as to maximize the learning signal for queries in $\mathcal{S}$ by introducing hard yet non-contradictory in-batch negatives.

5.1. Batch Selection Problem

Now, we formally define the batch hardness measure.

Definition 5.1 (Batch Hardness)**.** For a candidate subset $X \subset \mathcal{U}$, we define the batch hardness as

$$H(B) = \sum_{q_i \in \mathcal{S}} \max_{(q_j, d_j) \in \mathcal{S} \cup X} w_{ij}, \tag{7}$$

where $B = X \cup \mathcal{S}$, and w_{ij} is defined in Eq. 6.

The resulting batch selection problem is

$$\max_{X \subset \mathcal{U}} H(X \cup \mathcal{S}) \quad \text{subject to} \quad |X| = k. \tag{8}$$

where $k = b - s$.

Proposition 5.2 (NP-hardness of Batch Hardness Maximization)**.** *Maximizing the batch hardness objective in Eq. 8 is NP-hard.*

We prove this via reduction to the Maximum Coverage problem, with the proof deferred to Appendix A.3.3.

5.2. `HOBIT`'s Objective

`HOBIT` seeks a tractable batch selection objective that admits efficient optimization. The key idea is to replace the hard max in Eq. 7 using a smooth *Log-Sum-Exp* function:

$$\widetilde{h}_i(B) \;=\; \tau \log\left(\sum_{j \in B} \exp\left(\frac{w_{ij}}{\tau}\right)\right) \;;\; \text{for each } q_i \in \mathcal{S} \tag{9}$$

Definition 5.3 (`HOBIT`'s Objective)**.** The batch hardness score optimized by `HOBIT` is defined as the sum of the smooth marginal contributions for the queries in the seed set:

$$\widetilde{H}(B) = \sum_{q_i \in \mathcal{S}} \widetilde{h}_i(B) = \tau \sum_{q_i \in \mathcal{S}} \log\left(\sum_{j \in B} \exp\left(\frac{w_{ij}}{\tau}\right)\right). \tag{10}$$

The optimization objective is $\max_{X \subset \mathcal{U}} \widetilde{H}(\mathcal{S} \cup X)$ s.t. $|X| = b - s$.

Theorem 5.4 (Approximation Bounds)**.** *Let $B = \mathcal{S} \cup X$ be the full batch of size b with $|\mathcal{S}| = s$. For any real-valued weights w_{ij}, the `HOBIT`'s batch hardness $\widetilde{H}(B)$ bounds the hardness objective $H(B)$ in Eq. 7 as follows:*

$$H(B) \;\le\; \widetilde{H}(B) \;\le\; H(B) + s\tau \log b, \tag{11}$$

Proof deferred to Appendix A.4.1. □

Proposition 5.5 (Small-Temperature Regime)**.** *As $\tau \to 0^+$, the surrogate objective converges to the sum of maximum interactions over the seed set:*

$$\lim_{\tau \to 0^+} \widetilde{H}(B) = \sum_{q_i \in \mathcal{S}} \max_{j \in B} w_{ij} = H(B). \tag{12}$$

As a consequence, we have this corollary.

Corollary 5.6. *For any two batches B_1, B_2, the ranking between H and $\widetilde{H}$ is the same for $\tau \to 0^+$:*

$$\lim_{\tau \to 0^+} \left(\widetilde{H}(B_1) - \widetilde{H}(B_2)\right) = H(B_1) - H(B_2). \quad (13)$$

Proof deferred to Appendix A.4.2. □

Proposition 5.7 (Large-Temperature Regime)**.** *As $\tau \to \infty$, HOBIT's objective is asymptotically equivalent to:*

$$\lim_{\tau \to \infty} \widetilde{H}(B) \; = \; s\tau \log b + \mathcal{L}_{avg}(B). \quad (14)$$

where $\mathcal{L}_{avg}(B) = \sum_{q_i \in \mathcal{S}} \left(\frac{1}{b} \sum_{j \in B} w_{ij}\right)$

These results lead us to the following corollary.

Corollary 5.8. *For any two batches B_1 and B_2 of equal size, their ordering in the high-temperature limit is determined by the difference between their arithmetic-mean hardness scores:*

$$\lim_{\tau \to \infty} \left(\widetilde{H}(B_1) - \widetilde{H}(B_2)\right) = \sum_{q_i \in \mathcal{S}} \Big[\mathcal{L}_{avg}(B_1) - \mathcal{L}_{avg}(B_2)\Big]. \quad (15)$$

Proof deferred to Appendix A.4.3. □

Remark 5.9. Together, these two regimes show that $\widetilde{H}(B)$ adapts its selection behavior as a function of τ. As $\tau \to 0$, the optimization emphasizes the **maximum** interaction, whereas as $\tau \to \infty$, it favors the **average** interaction. This transition mirrors the behavior of the InfoNCE loss in Eq. 2: at low temperatures, the gradient is dominated by the hardest negative, while at high temperatures, contributions from all negatives become more uniform. *Accordingly, HOBIT ties the batch selection temperature τ in Eq. 10 to the same temperature used in the InfoNCE loss in Eq. 2.*

5.3. Submodularity and Greedy Optimization

We now show that the objective $\widetilde{H}(B = \mathcal{S} \cup X)$ is monotone and submodular in X.

Definition 5.10 (Marginal Gain)**.** For a set $M \subseteq \mathcal{U}$ and an element $v \in \mathcal{U} \setminus M$, the marginal gain of adding v to M is

$$\begin{aligned} \Delta(v \mid M, \mathcal{S}) &= \widetilde{H}(\mathcal{S} \cup M \cup \{v\}) - \widetilde{H}(\mathcal{S} \cup M) \\ &= \sum_{q_i \in \mathcal{S}} \tau \log\left(1 + \frac{\exp(w_{iv}/\tau)}{\sum_{j \in M} \exp(w_{ij}/\tau)}\right). \end{aligned} \quad (16)$$

Theorem 5.11 (Monotone submodularity)**.** *The objective $\widetilde{H}(\mathcal{S} \cup X)$ is monotone and submodular with respect to the set $X \subseteq \mathcal{U}$.*

Proof deferred to Appendix A.5. □

Corollary 5.12 (Greedy approximation guarantee)**.** *Under a cardinality constraint $|B| = b$, the greedy algorithm achieves a $(1 - 1/e)$-approximation to the optimal value of $\widetilde{H}(B)$. (Nemhauser et al., 1978)*

5.4. HOBIT's Algorithm

An epoch of dual-encoder training comprises $t = |D_{\text{trn}}|/b$ SGD steps. At a given training step k, applying HOBIT requires computing hardness scores w_{ij} (Eq. 6) using the current encoder parameters f_{θ^k}. In an ideal implementation, query and document embeddings would be recomputed at every SGD step, and each batch would be drawn from examples not yet encountered within the epoch. Such a strategy, however, is computationally impractical, as the cost of batch construction alone would dominate the overall training time.

To make batch selection tractable, we instead consider a class of *lagged batch construction* strategies, in which batches are formed using embeddings produced by a lagged encoder $f_{\theta^{k-\ell_k}}$ with delay $\ell_k \geq 0$. This abstraction enables a principled analysis of our algorithms: HOBIT and HOBIT-C as efficient, controlled approximations to the ideal step-wise batching implementation.

Algorithm 1 HOBIT Training

```
Input: Training pairs D_trn, epochs T, batch size b, seed set size s, non-contradiction weight α, temperature τ
Return: Trained dual encoder f_θ
for e = 1 to T do
   Compute embeddings {q_i, d_i} for HOBIT using f_θ^{e-1}, or use cached embeddings for HOBIT-C
   B^(e) ← ConstructBatch(D_trn, b, s, τ, α, {q_i, d_i})
   for each batch B ∈ B^(e) do
      Compute loss: L(B) ← InfoNCE(B, τ) (Eq. 2)
      θ ← GradDescent(L(B), θ)
   end for
end for
```

Our Algorithms. We present two practical instantiations of our framework that differ in how the lag ℓ_k is handled.

- **HOBIT.** For all SGD steps within epoch e, batch construction uses embeddings computed by the encoder at the end of the previous epoch. Consequently, for any training step k in epoch e, the lag satisfies $\ell_k \in \{0, 1, \ldots, t-1\}$.
- **HOBIT-C.** Batch construction relies on embeddings cached from forward passes performed during the earlier epoch $e - 1$. This results in a rolling lag in which different examples may be associated with embeddings computed at different parameter states.

In Appendix A.6, we provide formal approximation guarantees for both algorithms. Under a mild *step-wise drift*

Algorithm 2 ConstructBatch

Input: D_{trn}, b, s τ, α, $\{\vec{q}_i, \vec{d}_i\}$
Return: Ordered mini-batches $\mathcal{B}$
Initialize $\mathcal{U} \leftarrow D_{\text{trn}}$, $\mathcal{B} \leftarrow ()$
while $\mathcal{U} \neq \emptyset$ **do**
  Sample $\mathcal{S} \subseteq \mathcal{U}$ of size s uniformly; $\mathcal{U} \leftarrow \mathcal{U} \setminus \mathcal{S}$
  Set $X \leftarrow \emptyset$ and collect the candidate pool

$$\mathcal{C} \leftarrow \bigcup_{q_i \in \mathcal{S}} \text{TOP-}k\Big(q_i^\top d_j \mid (q_j, d_j) \in \mathcal{U}, k = 200\Big)$$

  while $|X| < b - s$ and $\mathcal{C} \neq \emptyset$ **do**
    Select $v^\star$ that yields the highest marginal gain;

$$v^\star = \operatorname*{argmax}_{v \in \mathcal{C}} \Delta(v \mid X, \mathcal{S}) \quad \text{(Eq. 16)}$$

    $X \leftarrow X \cup \{v^\star\}, \mathcal{C} \leftarrow \mathcal{C} \setminus \{v^\star\}$
  end while
  $B \leftarrow \mathcal{S} \cup X, \mathcal{U} \leftarrow \mathcal{U} \setminus X, \mathcal{B} \leftarrow \text{append}(\mathcal{B}, B)$
end while
return $\mathcal{B}$

assumption that: the embedding change per SGD step is bounded by a small ϵ, as is typical with Adam or SGD with small learning rates, the error from using lagged embeddings grows linearly with the lag ℓ_k. This highlights that in HOBIT, embeddings are synchronized at epoch boundaries, so the error depends on at most $\ell_k \leq t$; whereas in HOBIT-C, embeddings are drawn from a rolling cache spanning the previous epoch, yielding a worst-case error with $\ell_k \leq 2t$.

We provide the pseudocode for the dual encoder training using HOBIT in Alg. 1 and analyze its complexity next.

Complexity analysis. For constructing a batch of size b with seed set size s and candidate pool $\mathcal{C}$, lazy greedy maximization of the HOBIT's objective requires $\mathcal{O}(|\mathcal{C}| \log |\mathcal{C}|)$ marginal gain evaluations in the worst case. Each marginal gain computation costs $\mathcal{O}(s)$, as it aggregates contributions over the seed set. Therefore, the time complexity of lazy greedy batch construction is $\mathcal{O}(s|\mathcal{C}| \log |\mathcal{C}|)$.

6. Experiments and Results

Our experiments answer the following research questions:

RQ1 Does our principled, training loss-aware batching outperform other state-of-the-art batching methods?
RQ2 Does HOBIT deliver complementary gains when combined with existing hard-negative mining methods?
RQ3 How robust is HOBIT when q-d relevance labels are partially missing?
RQ4 What is the computational overhead introduced by HOBIT and HOBIT-C?
RQ5 How sensitive is HOBIT to τ, α, and s?

Table 1. Dataset statistics used in our experiments.

Dataset	Train Pairs	Dev Pairs	Test Pairs	Corpus Size
MS MARCO (Passage)	502,939	6,980	–	8,841,823
Natural Questions	58,880	8,757	3,610	21,015,324
TriviaQA	60,413	8,837	11,313	21,015,324

Table 2. Dual-encoder backbones used in our experiments.

Model	Layers	Hidden Dim.	# Parameters
roberta-base	12	768	125M
e5-large-unsupervised	24	1024	334M

6.1. Experimental Setup

Our experiments follow standard dense retrieval protocols from recent work (Karpukhin et al., 2020; Xiong et al., 2021; Qu et al., 2021; Monath et al., 2023). Models are trained with the InfoNCE loss using in-batch negatives, with most hyperparameters (e.g., batch size, τ) set to the defaults from prior studies.

Datasets. Following prior work (Monath et al., 2023; Xiong et al., 2021; Yang et al., 2024), we evaluate our method on three standard dense retrieval benchmarks: MS MARCO Passage Ranking (Nguyen et al., 2016), Natural Questions (NQ) (Kwiatkowski et al., 2019), and TriviaQA (Joshi et al., 2017). Table 1 summarizes the dataset statistics.

Encoders. Table 2 shows the two dual-encoder backbones: `roberta-base` (Liu et al., 2019), a standard MLM-pretrained encoder, and `e5-large-unsupervised` (Wang et al., 2022), pretrained with weak supervision for retrieval. Queries and documents are encoded with a shared encoder using [CLS] pooling and ℓ_2 normalization. For E5, we use task-specific prompts (e.g., `query:`, `passage:`).

Implementation Details. We train all models using InfoNCE loss (Eq. 2). We use a batch size of 128 and set the softmax temperature to $\tau = 0.05$, following prior work (Hou & Li, 2023; Lei et al., 2023). We optimize using AdamW (Loshchilov & Hutter, 2019) with a learning rate of 1×10^{-5}, weight decay 0.01, gradient clipping at 1.0, and a linear warmup of 800 steps. We train models for up to 40 epochs with early stopping based on development-set performance (patience of 5 epochs). We use mixed-precision (FP16) training throughout. For HOBIT, we set the seed set size $s = 8$. We use the same temperature for HOBIT as in the InfoNCE loss, i.e., $\tau = 0.05$, and α defaults to 1. All experiments allow up to 64 positives per query. We conduct

Table 3. Comparison of state-of-the-art batching methods across datasets and encoder backbones. **Bold** highlights the best-performing method, and underlining marks the second-best.

Method	MSMARCO Passage		NQ		TriviaQA	
	MRR@10	MRR@100	R@10	R@100	R@10	R@100
`roberta-base`						
Random	24.4	25.7	45.1	77.7	48.7	85.1
NGAME	26.8	28.1	47.9	78.9	51.5	86.9
BatchSampler	24.2	25.4	49.1	81.4	48.4	85.0
B3	23.4	24.7	44.4	76.1	46.9	82.9
HOBIT	**27.4**	**28.6**	**52.4**	**83.1**	**52.5**	**87.2**
`e5-large-unsupervised`						
Random	27.2	28.5	56.1	88.5	53.2	89.2
NGAME	28.6	29.9	61.3	**90.5**	56.0	90.1
BatchSampler	27.0	28.2	57.4	88.3	52.9	88.8
B3	27.0	28.3	60.4	90.0	52.5	87.9
HOBIT	**30.1**	**31.3**	**62.9**	89.9	**57.7**	**91.2**

Table 4. Impact of combining HOBIT with explicit negative mining. All experiments use `roberta-base` with 8 sampled negatives per query. We report MRR@10/100 for MSMARCO Passage and Recall@10/100 for NQ and TriviaQA.

Dataset	Method	MRR@10	MRR@100
MSMARCO	ANCE	**27.33**	**28.60**
	ANCE + HOBIT	27.29	28.53
	TriSampler	26.26	27.46
	TriSampler + HOBIT	**26.65**	**27.88**
		R@10	R@100
NQ	ANCE	**59.27**	84.63
	ANCE + HOBIT	56.43	**85.41**
	TriSampler	53.80	80.88
	TriSampler + HOBIT	**57.13**	**84.75**
		R@10	R@100
TriviaQA	ANCE	52.67	86.34
	ANCE + HOBIT	**54.46**	**88.34**
	TriSampler	53.76	85.60
	TriSampler + HOBIT	**54.11**	**87.75**

all experiments on a single NVIDIA B200 GPU.

Evaluation Metrics. At inference time, given a test query, the dual encoder retrieves relevant documents from the full corpus. We assess the performance using a comprehensive list of metrics: normalized Discounted Cumulative Gain@k (nDCG@k), Recall@k ($R@k$), and Mean Reciprocal Rank@k (MRR@k) for $k \in \{10, 100\}$. For brevity, we report $R@k$ for the NQ, TriviaQA datasets, and $MRR@k$ for MSMARCO, following prior work. We defer the complete results to Appendix A.9, with formal definitions provided in Appendix A.7.

6.2. Baselines

We group the baselines into two categories:

Batch Construction Methods. *Random* uniformly shuffles training instances at each epoch. *NGAME* performs clustering over the query embeddings to group semantically similar queries within the same batch (Dahiya et al., 2023). *BS* constructs batches via graph-based random walks over query-query similarity graphs (Yang et al., 2023). *B3* forms semantically coherent batches using teacher-guided representations and graph partitioning (Thirukovalluru et al., 2025). We use `embeddinggemma-300m` (Vera et al., 2025) as the teacher model for B3.

Hard-Negative Mining. We consider two representative hard-negative mining methods, ANCE (Xiong et al., 2021) and TriSampler (Yang et al., 2024), to assess whether applying HOBIT atop mined negatives to form batches that also contain rich in-batch negatives yields gains beyond training solely with the mined negatives under random batching.

Remark 6.1. Since the training set D_{trn} is small, HOBIT performs exact nearest neighbour search entirely on the GPU. For the much larger document corpus, hard negatives are efficiently mined using FAISS (Johnson et al., 2019).

6.3. RQ1: Comparison with Batching Methods

We report the results for RQ1 in Table 3 and highlight the following observations:

1. Random batching performs significantly worse, showing the value of having informative in-batch negatives.
2. B3 underperforms because it keeps batches fixed and fails to account for encoder drift during training. In contrast, NGAME derives batch order each epoch, ranking second-best. This underscores that effective batching must adapt continuously during training to remain synchronized with the encoder's representation space.
3. With E5-large, HOBIT outperforms prior methods by a significant margin in all settings, except for R@100 on NQ, where it trails NGAME by just 0.6 points.
4. Overall, HOBIT emerges as the best performer.

6.4. RQ2: Adding HOBIT to Hard-Negative Mining

Table 4 shows the results for RQ2.

1. *MSMARCO:* HOBIT yields limited gains when combined with ANCE and modest improvements with TriSampler, suggesting diminishing returns when strong mined negatives are already available.
2. *NQ:* While ANCE alone achieves the best Recall@10, combining it with HOBIT improves Recall@100, indicating benefits at larger cutoffs.
3. *TriviaQA:* HOBIT consistently improves performance when combined with both ANCE and TriSampler, achieving the best results across metrics.

Thus, when explicit hard negatives are available, enriching in-batch negatives either maintains performance or offers marginal gains, but with weaker negatives, HOBIT yields substantial gains. Therefore, HOBIT complements rather than replaces hard-negative mining.

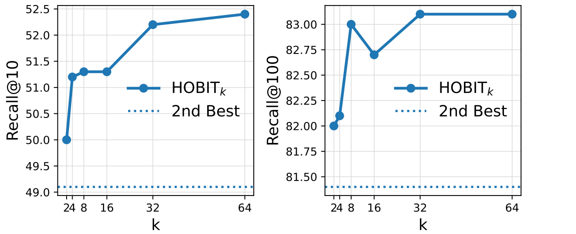

Figure 3. Effect of missing relevance labels on HOBIT$_k$ for NQ. Recall@10 (left) and Recall@100 (right) are shown, with dotted lines marking the strongest non-HOBIT batch sampler.

Table 5. Comparison of HOBIT vs. cached HOBIT-C. Efficiency gain shows per-epoch speedup of HOBIT-C over HOBIT. All settings use roberta-base encoder.

Dataset	Method	MRR@10	MRR@100	Efficiency
MSMARCO	HOBIT	27.4	28.6	–
	HOBIT-C	27.2	28.3	**14.3%**
		R@10	R@100	Efficiency
NQ	HOBIT	52.4	83.1	–
	HOBIT-C	51.7	82.4	**29.4%**
		R@10	R@100	Efficiency
TriviaQA	HOBIT	52.5	87.2	–
	HOBIT-C	52.2	86.9	**21.4%**

6.5. RQ3: Impact of Missing Labels on HOBIT

We evaluate HOBIT's robustness to sparse relevance supervision by limiting the number of positives per query ($k \in 2, 4, 8, 16, 32, 64$). All experiments use roberta-base. We excluded MSMARCO Passage since each query has only one ground-truth label. Results for NQ are shown in Fig. 3, and for TriviaQA in Fig. 5 in the Appendix.

1. HOBIT degrades gracefully as supervision becomes sparse and remains competitive even in the most restrictive setting ($k = 2$).
2. Performance improves monotonically with increasing k, and HOBIT consistently outperforms the strongest baseline for moderate sparsity levels ($k \geq 8$).
3. This robustness is attributed to the hardness scoring in HOBIT, where the non-contradiction term downweights false-negative like candidates during batch construction.

6.6. RQ4: Efficiency and Cached Batch Construction

We compare the computational efficiency of HOBIT with HOBIT-C in Table 5 that shows % efficiency gains.

1. HOBIT-C achieves substantial speedups across datasets (14–30%) with only minor drops in performance.
2. The performance drop remains under *one point* across all metrics, showing that caching robustly preserves the benefits of batch construction, which is also in line with Theorem A.5.
3. Overall, caching significantly reduces training time, making batch construction practical for large-scale datasets.

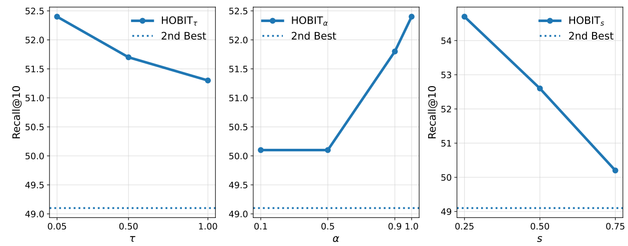

Figure 4. HOBIT's Sensitivity to τ, α, and s.

6.7. RQ5: Sensitivity to τ, α, and s

We study the sensitivity of HOBIT to three hyperparameters: temperature τ used in (Eq. 10), the mixing coefficient α used in w_{ij} (Eq. 6), and the seed set size s used to initialize each batch. We plot the results in Fig. 4.

1. **τ:** Performance peaks at the default $\tau = 0.05$ and declines gradually with higher τ, yet stays competitive with the strongest non-HOBIT sampler. This emphasizes the need to remain faithful to the underlying training loss when selecting negatives.
2. **α:** Performance rises steadily with α, peaking at 1.0, underscoring the importance of discarding near-positive hard negatives; a step overlooked by most baselines.
3. **s:** We vary the seed set size as a fraction of the batch and find that smaller seeds improve retrieval performance, as they give HOBIT more flexibility to enrich batches with diverse, informative hard negatives, whereas larger seeds limit the scope for batch refinement.

7. Conclusion

We introduced HOBIT, a principled approach to dual encoder training that constructs informative mini-batches. We identified two complementary criteria for an effective in-batch negative: it should be hard for the query and non-contradictory with the positive. We derived the batch order by optimizing a monotone submodular objective via a greedy algorithm with a $(1 - 1/e)$ guarantee. Our experiments on MSMARCO, NQ, and TriviaQA showed that HOBIT consistently outperformed state-of-the-art batch samplers, improving MRR and Recall by 1-3 points. For datasets with mined hard negatives, we showed that our method improved performance in some settings and maintained it in others, thereby complementing rather than replacing them. We proposed HOBIT-C as a scalable variant that reduced per-epoch training time by 12-30% with less than a 1-point performance drop. Overall, our work shows that having loss-aware mini-batches with informative in-batch negatives can substantially boost retrieval performance while adding minimal overhead, making them highly effective and practical for large-scale retrieval systems.

Impact Statement

This work introduces a lightweight, principled batching strategy for contrastive training of dense retrieval models, improving training stability and efficiency while reducing reliance on costly hard-negative mining. It is broadly applicable to retrieval and representation learning, and poses no direct societal risks.

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A. Appendix

A.1. Qualitative Analysis of In-Batch Negatives

To complement our theoretical analysis and quantitative evaluation, we present a qualitative example illustrating how different batch sampling strategies populate in-batch negatives for the same query–positive pair at different stages of training. The example is drawn from the MS MARCO Passage dataset and is intended to highlight systematic differences in the *type* of negatives selected by each method, rather than to provide exhaustive coverage.

We consider a fixed query and its associated positive passage, and inspect representative in-batch negatives produced by three batching strategies: (i) random batching, (ii) query–document similarity-based batching (e.g., ANCE-style), and (iii) HOBIT. For each method, we show examples corresponding to early training and to mid/late training, reflecting the evolution of the embedding space.

Table 6. Example of in-batch negatives selected by different batching strategies for a fixed MS MARCO query–positive pair at different stages of training.

Query–Positive Pair		
Query: When is the best time to travel to Iceland for the northern lights? **Positive Document:** In Iceland—especially far up in the north—it is possible to see the beautiful Aurora. The best time is from October to April each year, when we have clear skies and cold weather. It is ideal due to the fact that Iceland lies between two continents and has many excellent locations for viewing the Northern Lights.		
Batching Method	**Early Epoch In-Batch Negative**	**Mid / Late Epoch In-Batch Negative**
Random (R)	*Will dental insurance cover surgery* | Dental insurance policies typically cover procedures related to teeth and oral health. Medical insurance generally does not cover dental surgery, though coverage varies by policy.	*Federal court system definition* | A federal court is a court established by the authority of a federal government. The federal judiciary of the United States is responsible for interpreting and enforcing federal laws.
ANCE-style (A)	*Are the northern lights seasonal* | Prime seasons for viewing the northern lights are early autumn and spring, with visibility typically between September and March depending on solar activity.	*When do the northern lights appear in Iceland?* | The best time of year to see the Northern Lights in Iceland is between September and March, with clear skies and cold but dry conditions offering the best chance.
HOBIT (HOBIT)	*When can we see northern lights in Norway* | The Northern Lights usually appear between 6 p.m. and 1 a.m., with peak frequency around 10–11 p.m. during the dark months.	*How often do the northern lights appear* | The Northern Lights occur year round but are only visible when nights are dark. In regions such as Alaska, Canada, Norway, and Finland, they are most commonly seen from September through April.

This example illustrates that while ANCE-style batching increasingly selects negatives that are lexically and semantically very close to the positive passage, such negatives can be ambiguous and partially contradictory. In contrast, HOBIT consistently selects negatives that remain highly relevant to the query while avoiding near-duplicates of the positive, aligning with the design of the hardness score in Eq. 6.

A.2. Detailed Related Work

We provide a comprehensive review of the related literature on dual encoder training and hard-negative mining approaches below:

A.2.1. Dense retrieval and contrastive objectives.

Dual-encoder dense retrieval trained with contrastive objectives such as InfoNCE is the dominant paradigm for scalable retrieval (Karpukhin et al., 2020; van den Oord et al., 2018). In this setting, in-batch negatives serve as an efficient approximation to global negatives, but their effectiveness depends critically on batch composition (Karpukhin et al., 2020; Lin et al., 2021a; Wang & Isola, 2020), as the negative distribution in the softmax denominator directly shapes gradient signal and embedding geometry (Cao et al., 2022). Retrieval performance is further improved through better pre-training and objective design: domain-matched and corpus-aware pre-training yield stronger encoder initializations (Oguz et al., 2022; Gao & Callan, 2021; 2022; Wang et al., 2023; Shen et al., 2023), while weakly supervised and unsupervised contrastive objectives produce high-quality retrieval embeddings (Wang et al., 2022; Izacard et al., 2022; Gupta et al., 2024a). Several methods enhance robustness by modifying the contrastive loss while still relying on in-batch negatives (Li et al., 2021; Lu et al., 2022). To increase the effective number of negatives without prohibitive cost, momentum queues and memory banks reuse past embeddings (He et al., 2020; Chen et al., 2020), and retrieval-specific approximations such as negative caches and gradient accumulation improve convergence under hardware constraints (Lindgren et al., 2021; Kim et al., 2024; Lin et al., 2021b).

A.2.2. Hard-negative mining.

Hard-negative mining improves dense retrieval by retrieving challenging negatives from the corpus (Xiong et al., 2021; Yang et al., 2024; Lai et al., 2024; Shi et al., 2022; Chen et al., 2024; Monath et al., 2023). ANCE introduced asynchronous ANN-based mining, showing substantial gains over random or lexical BM25 negatives (Xiong et al., 2021), and subsequent systems expand the negative pool and reduce false negatives through cross-batch sharing and cross-encoder filtering (Qu et al., 2021; Ren et al., 2021b). Despite their effectiveness, these approaches are computationally expensive and prone to representation staleness due to delayed index updates (Xiong et al., 2021; Sun et al., 2022). Parallel work studies which negatives are most informative: SimANS shows that excessively hard negatives can introduce bias through false positives and proposes sampling ambiguous negatives (Zhou et al., 2022), while TriSampler enforces geometric constraints to avoid both trivial and false-negative-like samples (Yang et al., 2024). More broadly, noise-aware contrastive learning identifies false negatives as a central failure mode under sparse supervision (Chuang et al., 2020; Zhang & Stratos, 2021; Shi et al., 2023; Liu & Wang, 2023), motivating system-level remedies using auxiliary rankers and adversarial objectives (Zhang et al., 2022; Qu et al., 2021; Ren et al., 2021a).

A.3. Proofs from Problem Formulation

A.3.1. Proof of Lemma 4.1

Proof. Recall that the loss for the i-th item in the batch is

$$\mathcal{L}_i = -\frac{s_{ii}}{\tau} + \log\left(\sum_{j=1}^{b} \exp\left(\frac{s_{ij}}{\tau}\right)\right). \tag{17}$$

Since $s_{ij} = \vec{q_i}^\top \vec{d_j}$, we have $\nabla_{q_i} s_{ij} = \vec{d_j}$. Differentiating the two terms yields

$$\nabla_{\vec{q_i}}\left(-\frac{s_{ii}}{\tau}\right) = -\frac{1}{\tau}\vec{d_i}, \tag{18}$$

$$\nabla_{\vec{q_i}} \log\left(\sum_{j} \exp\left(\frac{s_{ij}}{\tau}\right)\right) = \frac{1}{\tau}\sum_{j} P_{ij}\,\vec{d_j}. \tag{19}$$

Combining the two expressions gives the result. □

A.3.2. Proof of Theorem 4.3

Proof. From the gradient expression for the InfoNCE loss,

$$\nabla_{\vec{q_i}}\mathcal{L}_i = \frac{1}{\tau}\left(\sum_{k=1}^{b} P_{ik}\,\vec{d_k} - \vec{d_i}\right). \tag{20}$$

We first expand the softmax-weighted document embedding:

$$\sum_{k=1}^{b} P_{ik}\,\vec{d_k} = (1-p-\epsilon)\vec{d_i} + p\vec{d_j} + \sum_{k\notin\{i,j\}} P_{ik}\,\vec{d_k}. \tag{21}$$

Subtracting $\vec{d_i}$ from both sides yields

$$\sum_{k=1}^{b} P_{ik}\,\vec{d_k} - \vec{d_i} = (1-p-\epsilon)\vec{d_i} - \vec{d_i} + p\vec{d_j} + \sum_{k\notin\{i,j\}} P_{ik}\,\vec{d_k} \tag{22}$$

$$= -(p+\epsilon)\vec{d_i} + p\vec{d_j} + \sum_{k\notin\{i,j\}} P_{ik}\,\vec{d_k}. \tag{23}$$

Now we regroup the terms to obtain:

$$-(p+\epsilon)\vec{d_i} + p\vec{d_j} + \sum_{k\notin\{i,j\}} P_{ik}\,\vec{d_k} \tag{24}$$

$$= p(\vec{d_j} - \vec{d_i}) + \sum_{k\notin\{i,j\}} P_{ik}(\vec{d_k} - \vec{d_i}), \tag{25}$$

where we used the identity

$$\sum_{k\notin\{i,j\}} P_{ik}\vec{d_i} = \epsilon\,\vec{d_i}. \tag{26}$$

Define

$$a = p(\vec{d_j} - \vec{d_i}), \qquad r = \sum_{k\notin\{i,j\}} P_{ik}(\vec{d_k} - \vec{d_i}). \tag{27}$$

Then

$$\sum_{k=1}^{b} P_{ik}\vec{d_k} - \vec{d_i} = a + r. \tag{28}$$

By the reverse triangle inequality,

$$\|a+r\|_2 \geq \|a\|_2 - \|r\|_2. \tag{29}$$

For the dominant term,

$$\|a\|_2 = p\|\vec{d_j} - \vec{d_i}\|_2 \tag{30}$$

$$= p\sqrt{\|\vec{d_j}\|_2^2 + \|\vec{d_i}\|_2^2 - 2\,\vec{d_i}^{\top}\vec{d_j}} \tag{31}$$

$$= p\sqrt{2(1-t_{ij})}, \tag{32}$$

where $t_{ij} = \vec{d_i}^{\top}\vec{d_j}$ and embeddings are unit-normalized.

For the residual term, using the triangle inequality and $\|\vec{d_k} - \vec{d_i}\|_2 \leq 2$,

$$\|r\|_2 \leq \sum_{k\notin\{i,j\}} P_{ik}\,\|\vec{d_k} - \vec{d_i}\|_2 \tag{33}$$

$$\leq 2\sum_{k\notin\{i,j\}} P_{ik} = 2\epsilon. \tag{34}$$

Combining the bounds,

$$\left\|\sum_{k=1}^{b} P_{ik}\vec{d_k} - \vec{d_i}\right\|_2 \geq p\sqrt{2(1-t_{ij})} - 2\epsilon. \tag{35}$$

Dividing by τ gives

$$\left\|\nabla_{\vec{q}_i}\mathcal{L}_i\right\|_2 \geq \frac{p}{\tau}\sqrt{2(1-t_{ij})} - \frac{2\epsilon}{\tau}, \tag{36}$$

which completes the proof. □

A.3.3. Proof of Proposition 5.2

Proof. We prove NP-hardness via a polynomial-time reduction from the classical Maximum Coverage problem, which is known to be NP-hard.

An instance of Maximum Coverage consists of a finite universe $U = \{u_1, \dots, u_n\}$, a collection of subsets $\mathcal{C} = \{S_1, \dots, S_m\}$ with $S_j \subseteq U$, and an integer k. The objective is to select k subsets whose union has maximum cardinality.

Given such an instance, we construct an instance of the batch selection problem as follows. For each element $u_i \in U$, we create a distinct seed example (q_i, d_i) and include it in the seed set $\mathcal{S}$. For each subset $S_j \in \mathcal{C}$, we create a distinct candidate example (q_j, d_j) and include it in the candidate pool $\mathcal{U}$.

We define the interaction weights w_{ij} synthetically to encode membership of elements in subsets:

$$w_{ij} = \begin{cases} 1, & \text{if } u_i \in S_j, \\ 0, & \text{otherwise.} \end{cases} \tag{37}$$

Additionally, we set all interactions among seed examples to zero, so that no seed example contributes hardness to another seed example.

Under this construction, for any subset $X \subset \mathcal{U}$, the inner maximization $\max_{(q_j,d_j)\in\mathcal{S}\cup X} w_{ij}$ evaluates to 1 if and only if there exists at least one candidate $(q_j, d_j) \in X$ such that $u_i \in S_j$, and evaluates to 0 otherwise. Consequently, the batch hardness reduces to

$$\begin{aligned} H(X) &= \sum_{q_i\in\mathcal{S}} \mathbb{I}[\exists (q_j, d_j) \in X \text{ such that } u_i \in S_j] \\ &= \left| \bigcup_{(q_j,d_j)\in X} S_j \right|. \end{aligned} \tag{38}$$

Thus, maximizing $H(X)$ subject to $|X| = k$ is equivalent to selecting k subsets from $\mathcal{C}$ whose union has maximum size, which is exactly the Maximum Coverage problem. Since Maximum Coverage is NP-hard, the batch hardness maximization problem is also NP-hard. □

A.4. Properties of the Surrogate Batch Hardness Objective

A.4.1. Approximation Bounds for Log-Sum-Exp

Proof. We analyze the bounds for a single query term $\widetilde{h}_i(B)$ for a fixed $q_i \in \mathcal{S}$ and then sum over the seed set. Let $w_i^\star = \max_{j\in B} w_{ij}$.

Lower bound. Since the exponential function maps real inputs to positive values, $\exp(w_{ij}/\tau) > 0$ for all j. The sum of positive terms is strictly greater than or equal to its largest term:

$$\sum_{j\in B} \exp\left(\frac{w_{ij}}{\tau}\right) \;\geq\; \max_{j\in B}\left[\exp\left(\frac{w_{ij}}{\tau}\right)\right] = \exp\left(\frac{w_i^\star}{\tau}\right). \tag{39}$$

Taking the logarithm (which is monotonic) and multiplying by $\tau > 0$:

$$\tau \log\left(\sum_{j\in B} \exp\left(\frac{w_{ij}}{\tau}\right)\right) \;\geq\; \tau \log\left(\exp\left(\frac{w_i^\star}{\tau}\right)\right) = w_i^\star. \tag{40}$$

Thus, $\widetilde{h}_i(B) \geq \max_{j\in B} w_{ij}$. Summing over all $q_i \in \mathcal{S}$ yields $\widetilde{H}(B) \geq H(B)$.

Upper bound. For every $j \in B$, we have $\exp(w_{ij}/\tau) \leq \exp(w_i^\star/\tau)$. We can bound the sum by replacing every term with the maximum term:

$$\sum_{j \in B} \exp\left(\frac{w_{ij}}{\tau}\right) \;\leq\; \sum_{j \in B} \exp\left(\frac{w_i^\star}{\tau}\right) = b \cdot \exp\left(\frac{w_i^\star}{\tau}\right). \tag{41}$$

Taking the logarithm and multiplying by τ:

$$\begin{aligned} \widetilde{h}_i(B) \;&\leq\; \tau \log\left(b \cdot \exp\left(\frac{w_i^\star}{\tau}\right)\right) && (42)\\ &=\; \tau\left(\log b + \frac{w_i^\star}{\tau}\right) && (43)\\ &=\; w_i^\star + \tau \log b. && (44) \end{aligned}$$

Summing over all $q_i \in \mathcal{S}$ yields:

$$\begin{aligned} \sum_{q_i \in \mathcal{S}} \widetilde{h}_i(B) \;&\leq\; \sum_{q_i \in \mathcal{S}} \left(\max_{j \in B} w_{ij} + \tau \log b\right) && (45)\\ &= H(B) + s\tau \log b. \end{aligned}$$

□

A.4.2. Small-Temperature Limit

Proof. Let $w_i^\star = \max_{j \in B} w_{ij}$. We analyze the term for a single query q_i:

$$\begin{aligned} \widetilde{h}_i(B) &= \tau \log\left(\sum_{j \in B} \exp\left(\frac{w_{ij}}{\tau}\right)\right) && (46)\\ &= \tau \log\left(\exp\left(\frac{w_i^\star}{\tau}\right) \sum_{j \in B} \exp\left(\frac{w_{ij} - w_i^\star}{\tau}\right)\right) && (47)\\ &= w_i^\star + \tau \log\left(\sum_{j \in B} \exp\left(\frac{w_{ij} - w_i^\star}{\tau}\right)\right). && (48) \end{aligned}$$

Let n_i be the number of items in B that achieve the maximum score $w_i^\star$. For any j where $w_{ij} < w_i^\star$, the term $\exp((w_{ij} - w_i^\star)/\tau)$ approaches 0 as $\tau \to 0^+$. The sum inside the logarithm, therefore, approaches n_i.

$$\lim_{\tau \to 0^+} \widetilde{h}_i(B) = w_i^\star + \lim_{\tau \to 0^+} \tau \log(n_i) = w_i^\star. \tag{49}$$

Summing over all $q_i \in \mathcal{S}$ yields the result. □

Corollary A.1. *For any two batches B_1, B_2, the ranking between H and $\widetilde{H}$ is constant for $\tau \to 0^+$:*

$$\lim_{\tau \to 0^+} \left(\widetilde{H}(B_1) - \widetilde{H}(B_2)\right) = H(B_1) - H(B_2). \tag{50}$$

Proof. Let $L = \lim_{\tau \to 0^+} (\widetilde{H}(B_1) - \widetilde{H}(B_2))$. By the linearity of limits:

$$L = \sum_{q_i \in \mathcal{S}} \lim_{\tau \to 0^+} \widetilde{h}_i(B_1) - \sum_{q_i \in \mathcal{S}} \lim_{\tau \to 0^+} \widetilde{h}_i(B_2). \tag{51}$$

Substituting the pointwise limit from Proposition 5.5 yields the stated result. □

A.4.3. Large-Temperature Limit

Proof. We use the Taylor expansions $e^x = 1 + x + \mathcal{O}(x^2)$ and $\log(1+x) = x + \mathcal{O}(x^2)$ for small x. As $\tau \to \infty$, $w_{ij}/\tau \to 0$. Analyzing the inner sum for a single query q_i:

$$\sum_{j \in B} \exp\left(\frac{w_{ij}}{\tau}\right) = \sum_{j \in B} \left(1 + \frac{w_{ij}}{\tau} + \mathcal{O}(\tau^{-2})\right) \tag{52}$$

$$= b + \frac{1}{\tau} \sum_{j \in B} w_{ij} + \mathcal{O}(\tau^{-2}) \tag{53}$$

$$= b \left(1 + \frac{1}{b\tau} \sum_{j \in B} w_{ij} + \mathcal{O}(\tau^{-2})\right). \tag{54}$$

Substituting this back into the expression for $\widetilde{h}_i(B)$:

$$\widetilde{h}_i(B) = \tau \log \left[b \left(1 + \frac{1}{b\tau} \sum_{j \in B} w_{ij}\right)\right] \tag{55}$$

$$= \tau \log b + \tau \log \left(1 + \frac{1}{b\tau} \sum_{j \in B} w_{ij}\right) \tag{56}$$

$$= \tau \log b + \tau \left(\frac{1}{b\tau} \sum_{j \in B} w_{ij} + \mathcal{O}(\tau^{-2})\right) \tag{57}$$

$$= \tau \log b + \frac{1}{b} \sum_{j \in B} w_{ij} + \mathcal{O}(\tau^{-1}). \tag{58}$$

Summing over all $q_i \in \mathcal{S}$ yields the proposition. □

Corollary A.2. *For any two batches B_1, B_2 of equal size $|B_1| = |B_2| = b$, the ranking in the high-temperature limit is determined by the difference of the arithmetic mean of hardness scores:*

$$\lim_{\tau \to \infty} \left(\widetilde{H}(B_1) - \widetilde{H}(B_2)\right) = \sum_{q_i \in \mathcal{S}} \mathcal{L}_{avg}(B_1) \; - \; \mathcal{L}_{avg}(B_2). \tag{59}$$

where $\mathcal{L}_{avg}(B) = \sum_{q_i \in \mathcal{S}} \frac{1}{b} \sum_{j \in B} w_{ij}$.

Proof. Using the expansion from Proposition 5.7, we write the difference:

$$\widetilde{H}(B_1) - \widetilde{H}(B_2) \approx \\ (|\mathcal{S}|\tau \log b + \mathcal{L}_{\text{avg}}(B_1)) - (|\mathcal{S}|\tau \log b + \mathcal{L}_{\text{avg}}(B_2)) \tag{60}$$

The divergent term $|\mathcal{S}|\tau \log b$ cancels out exactly because the batch sizes are equal. Taking the limit $\tau \to \infty$ removes the $\mathcal{O}(\tau^{-1})$ remainder, leaving only the difference of the means. □

A.5. HOBIT: Proof of Monotone Submodularity

Proof. We first show monotonicity. For any $N \subseteq \mathcal{U}$ and any $v \in \mathcal{U} \setminus N$, adding v increases the argument of the logarithm for each $q_i \in \mathcal{S}$, since all exponential terms are non-negative. Therefore, $\widetilde{H}(\mathcal{S} \cup N \cup \{v\}) \geq \widetilde{H}(\mathcal{S} \cup N)$, and the objective is monotone.

We now prove submodularity via the diminishing-returns property. Consider two sets $M \subseteq N \subseteq \mathcal{U}$ and an element $v \in \mathcal{U} \setminus N$. For a fixed query $q_i \in \mathcal{S}$, define

$$Z_i(M) = \sum_{j \in M} \exp(w_{ij}/\tau), \qquad Z_i(N) = \sum_{j \in N} \exp(w_{ij}/\tau).$$

Since $M \subseteq N$, we have $Z_i(N) \geq Z_i(M)$.

From Eq. (16), the per-query marginal contribution of v when added to a set M is

$$\Delta_i(v \mid M) = \tau \log\left(1 + \frac{\exp(w_{iv}/\tau)}{Z_i(M)}\right).$$

The function $g(x) = \tau \log(1 + c/x)$ for fixed $c > 0$ is non-increasing in $x > 0$. Therefore, increasing the denominator from $Z_i(M)$ to $Z_i(N)$ can only decrease the marginal gain, implying

$$\Delta_i(v \mid M) \geq \Delta_i(v \mid N).$$

Summing this inequality over all $q_i \in \mathcal{S}$ yields

$$\Delta(v \mid M, \mathcal{S}) = \sum_{q_i \in \mathcal{S}} \Delta_i(v \mid M) \;\geq\; \sum_{q_i \in \mathcal{S}} \Delta_i(v \mid N) = \Delta(v \mid N),$$

which establishes the diminishing-returns property. Hence, $\widetilde{H}(N)$ is submodular. □

A.6. Lagged Batch Construction: Formal Guarantees

In this appendix, we provide formal guarantees for lagged batch construction. we introduce a *step-wise drift assumption*, and derive how this drift accumulates over the lag period ℓ_k, explicitly linking the lag duration to the approximation bounds.

A.6.1. Step-wise Embedding Drift

We assume that the embedding parameters change smoothly at each SGD step. This is consistent with standard optimization (e.g., SGD, Adam), where the learning rate limits the magnitude of parameter updates per step.

Assumption A.3 (Bounded step-wise drift)**.** There exists a constant $\epsilon > 0$ such that for any optimization step k and all indices i, the change in embeddings between consecutive steps is bounded by:

$$\|\vec{q}_i^{\,(k)} - \vec{q}_i^{\,(k-1)}\|_2 \leq \epsilon, \qquad \|\vec{d}_i^{\,(k)} - \vec{d}_i^{\,(k-1)}\|_2 \leq \epsilon. \tag{61}$$

A.6.2. Accumulated Drift Under Lag

We first establish that the total embedding drift for a lag of ℓ_k is bounded linearly by the number of lagged steps.

Lemma A.4 (Lag-dependent drift bound)**.** *Under Assumption A.3, for a current step k and a lag $\ell_k \geq 0$, the total embedding drift is bounded by:*

$$\|\vec{q}_i^{\,(k)} - \vec{q}_i^{\,(k-\ell_k)}\|_2 \leq \ell_k \epsilon, \qquad \|\vec{d}_i^{\,(k)} - \vec{d}_i^{\,(k-\ell_k)}\|_2 \leq \ell_k \epsilon. \tag{62}$$

Proof. We express the difference between the current and lagged embedding as a telescoping sum of single-step updates. Using the triangle inequality and Assumption A.3:

$$\begin{aligned}
\|\vec{q}_i^{\,(k)} - \vec{q}_i^{\,(k-\ell_k)}\|_2 &= \left\| \sum_{j=0}^{\ell_k - 1} \left(\vec{q}_i^{\,(k-j)} - \vec{q}_i^{\,(k-j-1)} \right) \right\|_2 \\
&\leq \sum_{j=0}^{\ell_k - 1} \left\| \vec{q}_i^{\,(k-j)} - \vec{q}_i^{\,(k-j-1)} \right\|_2 \\
&\leq \sum_{j=0}^{\ell_k - 1} \epsilon \;=\; \ell_k \epsilon.
\end{aligned} \tag{63}$$

The derivation for $\vec{d}_i$ is identical. □

A.6.3. Stability Under Lag

Lemma A.5 (Lag-dependent perturbation)**.** *Under Assumption A.3, for any fixed batch B and lag ℓ_k,*

$$\left|\widetilde{H}_{\theta_k}(B) - \widetilde{H}_{\theta_{k-\ell_k}}(B)\right| \;\leq\; b(2+2\alpha)\ell_k\epsilon. \tag{64}$$

Proof. First, we bound the drift in the hardness scores w_{ij}. For unit-norm embeddings, the difference in dot products satisfies $|u^\top v - u'^\top v'| \leq \|u-u'\| + \|v-v'\|$. Combining this with Lemma A.4:

$$|w_{ij}^{(k)} - w_{ij}^{(k-\ell_k)}| \leq 2\ell_k\epsilon + 2\alpha\ell_k\epsilon \;=\; (2+2\alpha)\ell_k\epsilon. \tag{65}$$

The function $\tau \log \sum_{j\in B} \exp(w_{ij}/\tau)$ is 1-Lipschitz with respect to the scores w_{ij}. Summing over the batch of size b yields the stated bound. □

A.6.4. Greedy Optimization Under Lag

Theorem A.6 (Greedy guarantee with lag-dependent error)**.** *Let $B_k^\star$ be an optimal batch for $\widetilde{H}_{\theta_k}$, and let $\widehat{B}_k$ be the batch obtained by greedy maximization of the lagged objective $\widetilde{H}_{\theta_{k-\ell_k}}$. Under Assumption A.3,*

$$\widetilde{H}_{\theta_k}(\widehat{B}_k) \;\geq\; (1-1/e)\,\widetilde{H}_{\theta_k}(B_k^\star) - 2b(2+2\alpha)\ell_k\epsilon. \tag{66}$$

Proof. By the standard greedy guarantee for monotone submodular maximization (Corollary 5.12) on the lagged objective:

$$\widetilde{H}_{\theta_{k-\ell_k}}(\widehat{B}_k) \geq (1-1/e)\,\widetilde{H}_{\theta_{k-\ell_k}}(B_k^\star). \tag{67}$$

Applying Lemma A.5 to relate the lagged objective back to the current parameters θ_k:

$$\begin{aligned}
\widetilde{H}_{\theta_k}(\widehat{B}_k) &\geq \widetilde{H}_{\theta_{k-\ell_k}}(\widehat{B}_k) - b(2+2\alpha)\ell_k\epsilon \\
&\geq (1-1/e)\,\widetilde{H}_{\theta_{k-\ell_k}}(B_k^\star) - b(2+2\alpha)\ell_k\epsilon \\
&\geq (1-1/e)\left(\widetilde{H}_{\theta_k}(B_k^\star) - b(2+2\alpha)\ell_k\epsilon\right) - b(2+2\alpha)\ell_k\epsilon \\
&= (1-1/e)\,\widetilde{H}_{\theta_k}(B_k^\star) - \underbrace{\Big((1-1/e)+1\Big)}_{<2} b(2+2\alpha)\ell_k\epsilon \\
&\geq (1-1/e)\,\widetilde{H}_{\theta_k}(B_k^\star) - 2b(2+2\alpha)\ell_k\epsilon.
\end{aligned} \tag{68}$$

□

Remark A.7. We make the following two remarks:

- For **HOBIT**, where embeddings are fixed to the state at the end of the previous epoch, the lag grows linearly with the step count within the current epoch, bounded by $\ell_k \leq t$. Here, $t = \frac{|D_{\text{trn}}|}{b}$ denotes the number of gradient steps in an epoch of training the dual encoder.
- For **HOBIT-C**, where embeddings are retrieved from a rolling cache populated during the previous epoch, an embedding used at step k may originate from any point in the previous epoch. In the worst case, the lag is bounded by $\ell_k \leq 2t$.

A.7. Evaluation Metrics

We evaluate retrieval performance using Mean Reciprocal Rank@k (MRR@k), normalized Discounted Cumulative Gain@k (nDCG@k), Precision@k (P@k), and Recall@k (R@k), following standard practice in dense retrieval (Monath et al., 2023; Yang et al., 2024). Metrics are reported for $k \in \{10, 100\}$.

Let $\mathcal{Q}^{\text{tst}}$ denote the test query set with $n = |\mathcal{Q}^{\text{tst}}|$ queries. For a query $q \in \mathcal{Q}^{\text{tst}}$, let $\mathcal{R}_q^k$ be the ranked list of the top-k retrieved documents from the corpus D, and let $\mathcal{G}_q$ denote the set of relevant documents for q.

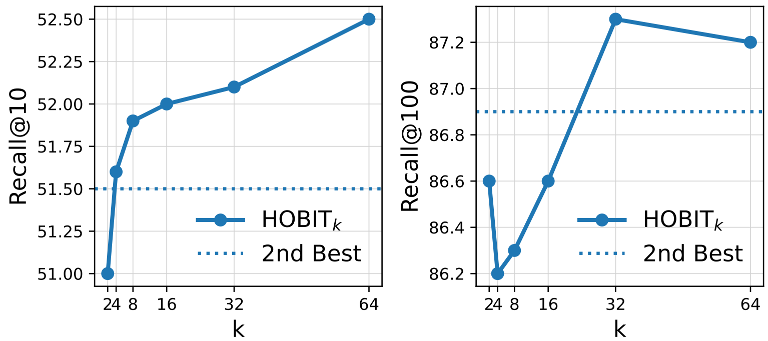

Figure 5. Effect of missing relevance labels on HOBIT$_k$ for TriviaQA. Recall@10 (left) and Recall@100 (right) are shown, with dotted lines marking the strongest non-HOBIT batch sampler.

Mean Reciprocal Rank.

$$\mathrm{MRR}@k = \frac{1}{n} \sum_{q \in \mathcal{Q}^{\mathrm{tst}}} \frac{1}{\min \left\{ \mathrm{rank}(d) \mid d \in \mathcal{R}_q^k \cap \mathcal{G}_q \right\}} \tag{69}$$

where the reciprocal rank is defined to be zero if no relevant document appears within the top-k retrieved results.

Precision and Recall.

$$\mathrm{P}@k = \frac{1}{n} \sum_{q \in \mathcal{Q}^{\mathrm{tst}}} \frac{\left| \mathcal{R}_q^k \cap \mathcal{G}_q \right|}{k}, \tag{70}$$

$$\mathrm{R}@k = \frac{1}{n} \sum_{q \in \mathcal{Q}^{\mathrm{tst}}} \frac{\left| \mathcal{R}_q^k \cap \mathcal{G}_q \right|}{|\mathcal{G}_q|}. \tag{71}$$

Normalized Discounted Cumulative Gain. For a query q, DCG@k is defined as

$$\mathrm{DCG}_q@k = \sum_{i=1}^{k} \frac{\mathbb{I}[d_i \in \mathcal{G}_q]}{\log_2(i+1)}, \tag{72}$$

where d_i is the document ranked at position i. The ideal DCG, IDCG$_q$@k, is computed by ranking all relevant documents for q at the top. The normalized metric is then

$$\mathrm{nDCG}@k = \frac{1}{n} \sum_{q \in \mathcal{Q}^{\mathrm{tst}}} \frac{\mathrm{DCG}_q@k}{\mathrm{IDCG}_q@k}. \tag{73}$$

A.8. Effect of Missing Labels on TriviaQA

A.9. Extended Evaluation

In this appendix, we report extended retrieval metrics for all batch construction methods across datasets and encoder architectures. While the main paper focuses on primary metrics (MRR@10 for MSMARCO and Recall@10/100 for NQ and TriviaQA), these tables provide a more comprehensive view of retrieval behaviour across ranking depths.

For each dataset, we report nDCG@k, Recall@k, and MRR@k for $k \in \{1, 5, 10, 20, 100\}$. All results follow the same training and evaluation protocol described in Section 6.1. For clarity, we group results by encoder and highlight HOBIT in blue.

Table 7. Extended retrieval metrics on MSMARCO Passage. We report nDCG@k, Recall@k, and MRR@k for $k \in \{1, 5, 10, 20, 100\}$ across batch construction methods and encoders. Results are grouped by encoder, with HOBIT highlighted.

Method	nDCG@1	nDCG@5	nDCG@10	nDCG@20	nDCG@100	R@1	R@5	R@10	R@20	R@100	MRR@1	MRR@5	MRR@10	MRR@20	MRR@100
roberta-base															
Random	14.7	27.2	30.8	33.7	37.4	14.3	38.8	49.9	61.2	80.4	14.7	23.7	24.4	25.0	25.7
NGAME	15.8	28.8	32.8	35.6	39.2	15.3	40.8	52.8	63.8	83.1	15.8	25.2	26.8	27.6	28.1
BatchSampler	16.2	29.2	33.0	35.9	39.5	15.7	41.4	52.9	64.1	**83.2**	16.2	23.9	24.2	25.2	25.4
B3	13.9	25.2	28.6	31.5	35.4	13.5	35.8	46.3	57.4	78.1	13.9	22.0	23.4	24.2	24.7
HOBIT	**16.3**	**29.4**	**33.4**	**36.2**	**39.6**	**15.7**	**41.6**	**53.7**	**64.6**	82.8	**16.3**	**25.8**	**27.4**	**28.2**	**28.6**
e5-large-unsupervised															
Random	15.5	29.6	33.7	36.6	40.2	15.1	42.8	55.4	66.7	85.3	15.5	25.5	27.2	28.0	28.5
NGAME	16.4	30.9	35.3	38.3	41.7	16.0	44.2	57.4	69.1	87.2	16.4	26.8	28.6	29.5	29.9
BatchSampler	15.0	29.3	33.5	36.3	39.9	14.5	42.7	55.5	66.2	85.3	15.0	25.2	27.0	27.7	28.2
B3	15.3	29.4	33.4	36.3	39.9	14.9	42.5	54.6	65.9	84.8	15.3	25.4	27.0	27.8	28.3
HOBIT	**18.1**	**32.3**	**36.6**	**39.7**	**43.0**	**17.6**	**45.5**	**58.6**	**70.4**	**87.8**	**18.1**	**28.3**	**30.1**	**30.9**	**31.3**

Table 8. Extended retrieval metrics on Natural Questions (NQ). We report nDCG@k, Recall@k, and MRR@k for $k \in \{1, 5, 10, 20, 100\}$ across batch construction methods and encoders. Results are grouped by encoder, with HOBIT highlighted.

Method	nDCG@1	nDCG@5	nDCG@10	nDCG@20	nDCG@100	R@1	R@5	R@10	R@20	R@100	MRR@1	MRR@5	MRR@10	MRR@20	MRR@100
roberta-base															
Random	13.2	23.9	27.7	30.8	34.9	11.6	33.9	45.1	56.9	77.8	13.2	21.7	23.2	24.0	24.6
NGAME	15.3	25.6	29.8	32.8	36.7	13.5	35.5	47.9	59.2	78.9	15.3	23.7	25.4	26.2	26.7
BatchSampler	15.7	26.5	30.7	33.9	37.9	13.9	36.6	49.1	61.3	81.5	15.7	24.4	26.1	26.9	27.4
B3	13.5	23.0	27.2	30.2	34.2	11.9	32.2	44.5	55.8	76.1	13.5	21.2	22.9	23.7	24.2
HOBIT	**16.6**	**28.3**	**32.7**	**35.8**	**39.6**	**14.9**	**39.3**	**52.4**	**63.7**	**83.2**	**16.6**	**25.9**	**27.6**	**28.4**	**28.9**
e5-large-unsupervised															
Random	18.0	31.0	35.3	38.9	42.7	15.7	43.3	56.1	69.6	88.5	18.0	28.4	30.2	31.1	31.5
NGAME	20.4	34.4	39.1	42.3	45.7	17.9	47.6	61.3	73.5	**90.5**	20.4	31.6	33.4	34.2	34.6
BatchSampler	17.6	31.0	35.6	39.0	42.7	15.6	43.7	57.4	70.0	88.3	17.6	28.1	30.0	30.9	31.3
B3	18.8	33.0	37.9	41.2	44.7	16.6	46.1	60.4	72.5	90.0	18.8	30.1	32.1	32.9	33.3
HOBIT	**22.8**	**36.5**	**41.1**	**44.2**	**47.3**	**20.0**	**49.5**	**62.9**	**74.5**	89.9	**22.8**	**34.0**	**35.7**	**36.5**	**36.8**

Table 9. Extended retrieval metrics on TriviaQA. We report nDCG@k, Recall@k, and MRR@k for $k \in \{1, 5, 10, 20, 100\}$ across batch construction methods and encoders. Results are grouped by encoder, with HOBIT highlighted.

Method	nDCG@1	nDCG@5	nDCG@10	nDCG@20	nDCG@100	R@1	R@5	R@10	R@20	R@100	MRR@1	MRR@5	MRR@10	MRR@20	MRR@100
roberta-base															
Random	63.2	59.9	59.7	61.2	67.8	12.0	35.3	48.7	62.2	85.1	63.2	71.2	71.8	72.1	72.2
NGAME	66.6	63.1	62.9	64.6	70.7	12.9	37.5	51.5	65.5	86.9	66.6	74.1	74.7	75.0	75.1
BatchSampler	63.4	59.6	59.4	61.0	67.6	12.1	35.0	48.4	61.9	85.0	63.4	71.0	71.8	72.0	72.1
B3	62.1	58.4	58.0	59.3	65.8	11.7	34.0	46.9	60.0	82.9	62.1	69.9	70.6	70.9	71.0
HOBIT	**67.4**	**64.2**	**64.0**	**65.5**	**71.4**	**13.2**	**38.5**	**52.5**	**66.2**	**87.2**	**67.4**	**75.0**	**75.5**	**75.7**	**75.8**
e5-large-unsupervised															
Random	67.4	64.7	64.5	66.1	72.4	13.0	38.6	53.2	67.1	89.2	67.4	74.8	75.5	75.7	75.8
NGAME	71.0	68.0	68.0	69.8	75.5	14.3	41.6	56.0	70.8	90.1	71.0	77.9	78.5	78.7	78.7
BatchSampler	67.7	64.2	64.1	65.7	72.0	13.0	38.6	52.9	66.7	88.8	67.7	74.7	75.3	75.5	75.6
B3	67.4	64.0	63.7	65.1	71.4	13.1	38.5	52.5	65.9	87.9	67.4	74.6	75.2	75.5	75.6
HOBIT	**72.8**	**69.6**	**69.7**	**71.3**	**76.7**	**14.7**	**42.5**	**57.7**	**72.0**	**91.2**	**72.8**	**79.5**	**80.0**	**80.2**	**80.2**